Typed Judgment, Deterministic Enforcement: A Consent-Gated Gateway Architecture for Inter-Agent Collaboration over Messaging Channels

Yizhang Cai, Jun Liu, Di Ma, Ke Yu, Ke Li

School of Artificial Intelligence, Beijing University of Posts and Telecommunications
Beijing Guoshu Zhilian Technology Co., Ltd.
2025111259@bupt.cn

The boundary we study

Messaging events are unstructured and may be duplicated, delayed, or replayed; agent protocols expect authenticated endpoints, structured requests, and task state.

- Free-text delegation lets model output become an invocation.
- Static rules cannot resolve natural-language ambiguity.

Design rule. The model answers closed questions; policy code owns thresholds and effects.

Seven modules, four invariants

M1 ingress authentication
M2 caller / conversation isolation
M3 discovery and routing
M4 consent and policy
M5 identity-bound execution
M6 state, idempotency, recovery
M7 delivery and telemetry

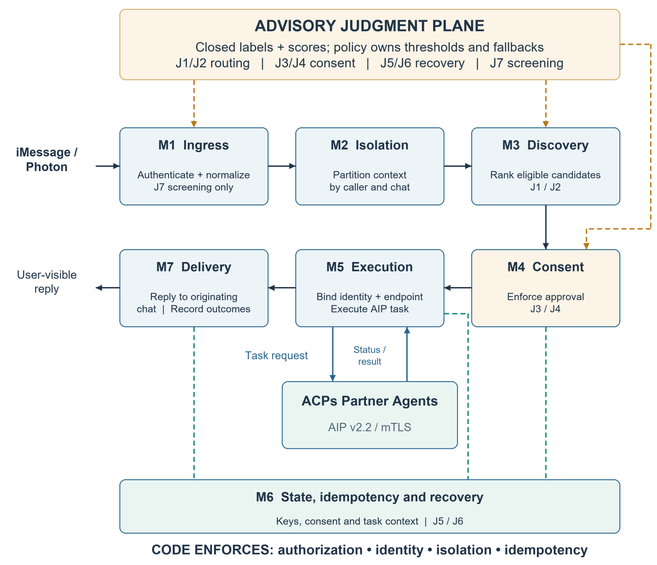
In code: authorization • identity • isolation • idempotency

The judgment plane: seven closed questions

The layer returns closed labels or probabilities only; it never emits free text, never calls a tool, and only parameterizes policy thresholds.

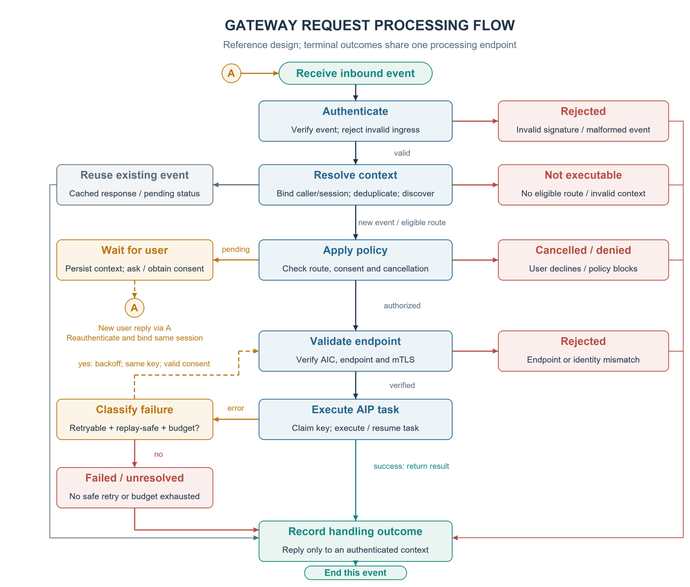
J1 intent **J2** routing
J3 ambiguity **J4** consent
J5 replay **J6** retryability
J7 injection

Architecture



M1 ingress, M2 partition by caller and conversation, M3 discovery, M4 consent, M5 identity-bound execution, M6 state, M7 delivery.

Processing flow



Deduplication precedes every model call; a pending consent re-enters at authentication; a failed identity or endpoint check never reaches execution.

Results: routing (130 requests, 3 deliveries)

Direct	48/48/50 of 50
Ambiguous	39/38/38 of 40
Out of scope	0 of 40

Results: invariants (paired ablations)

Arm	Control	Removed
Identity binding	0/30	30/30
Caller isolation	0/30	30/30
Idempotency	30 exec.	60 exec.

Same-code controls; each arm changes exactly one check.

Consent and judgment

Consent gate: **0**/**40** before consent, **36**/**36** exactly one action; replayed consents add nothing.

	Typed	Free text
Accuracy	94.1%	83.9%
Brier	0.095	0.266
ECE	0.026	0.144

Takeaway

Model advises; code controls identity, state, and every side effect.

Consent **0**/**40** before, **36**/**36** after. Judgment **94.1%** vs **83.9%**; Brier **0.095** vs **0.266**.

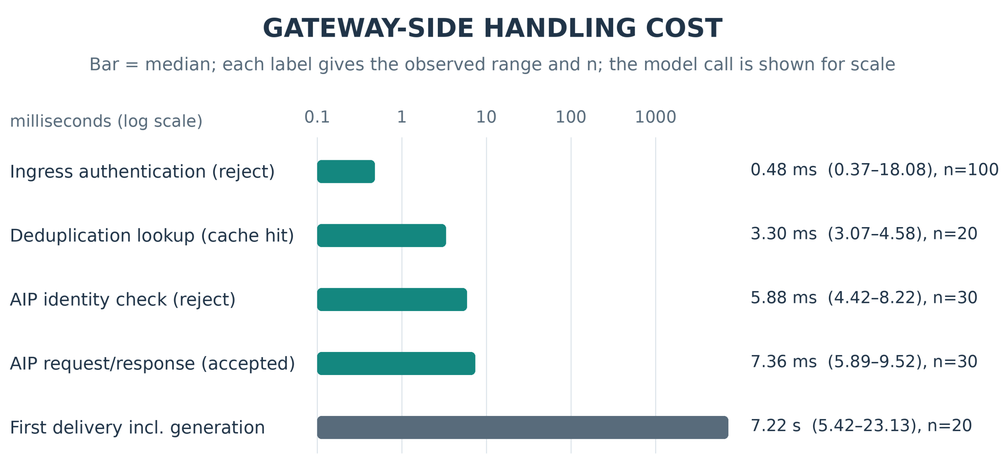
Artifact: `github.com/ankelas/consent-gated-gateway-artifact`

S&T Program of Xiongan New Area (No. XA2026111001K).

Injection stress set (200 injected, 256 benign near-misses)

Both answer shapes catch every injected item (0% miss), yet flag about half of the quoted benign mentions: 52.5% typed and 59.6% free text, which is why the injection score only parameterizes policy.

Gateway cost



Each gateway operation is milliseconds; an end-to-end delivery is dominated by assistant generation.

LaTeX TikZposter